

HEALTH-BAND Neuro: Patent Process and Timeline

Inventor: Srikanth Patchava **Assignee:** Embedded Operating Systems Research Foundation (EoS) **Core Patent:** US Provisional App. No. ⁶⁴/073,334 (Filed May 23, 2026)

This document outlines the strategic roadmap for securing intellectual property rights for the HEALTH-BAND Neuro, detailing the transition from provisional filing to non-provisional status, and the execution of the Continuation-In-Part (CIP) strategy.

Phase 1: Provisional Patent (Current Status)

The provisional patent establishes the priority date and secures the core architecture of the HEALTH-BAND Neuro.

- **Filing Date:** May 23, 2026
- **Expiration Date:** May 23, 2027 (12-month window)
- **Core Claims Protected:**
 - **Zero-Hole Architecture:** The structural design utilizing a dual-purpose USB-C clasp mechanism (male plug and female receptacle) to eliminate dedicated charging ports and sensor windows.
 - **Integrated Breath Channel:** The specific integration of a micro-perforated grille and hydrophobic membrane surrounding the USB-C female port to facilitate BAC and VOC sensing.
 - **Bidirectional Neuromuscular Interface:** The use of shared platinum electrodes on the inner band surface for both sEMG input (gesture control) and TENS output (neuromodulation therapy).

Action Items (Phase 1):

- ☑ File Provisional Application.
- ☑ Generate engineering block diagrams and cross-sections (Fig 1-3).

- ☐ Draft full non-provisional claims based on prototype testing.
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Phase 2: Non-Provisional Utility Patent (Months 6-10)

Before the provisional expires, a full non-provisional utility patent application must be filed to begin the formal examination process.

- **Target Filing Date:** November 2026 - March 2027
 - **Required Documentation:**
 - Formal claim set (independent and dependent claims).
 - Detailed description of the preferred embodiment.
 - Formal USPTO-compliant drawings.
 - **Strategy:** Emphasize the mechanical and electrical synergy of the clasp mechanism, specifically how the pass-through charging operates seamlessly while the device is unclashed.
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Phase 3: Continuation-In-Part (CIP) Strategy (Months 10-12)

The CIP application allows for the addition of new material that was not present in the original provisional filing, while maintaining the original priority date for the foundational claims.

- **Target Filing Date:** April 2027 - May 23, 2027
- **New Claims to be Added (CIP):**
 1. **Holographic Micro-LED Display:** The integration of a micro-LED array positioned behind a holographic diffuser film embedded flush in the outer surface of the flexible wristband.
 2. **Bezel-Less Integration:** The specific manufacturing process allowing the display area to be visually indistinguishable from the rest of the band when inactive.

Action Items (Phase 3):

- ☒ Generate CIP concept diagrams.
 - ☒ Draft specific CIP claim language (completed in HEALTH_BAND_Neuro_CIP_Strategy.md).
 - ☐ File CIP alongside or shortly after the non-provisional utility application.
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Phase 4: Patent Prosecution and Issuance (Years 1-3)

Following the non-provisional and CIP filings, the applications enter the examination phase.

- **Office Actions:** Respond to USPTO examiner rejections or objections.
- **Target Issuance:** 2028 - 2029.



HEALTH-BAND Neuro Patent Overview